Day 5 Evidence Workbooklet

Boolean Combinations and Week 1 Synthesis

Engineering practice to separate evidence to Boolean expression to go/no-go reasoning to support route

Student copy. Guided workbooklet, not the mastery quiz. Print format: duplex, left-stapled packet.

Name		**Section**		**Date**	
Score	______/40		Mastery check	secure / developing / needs support	

Concrete engineering situation

The pump-flow team now has three kinds of evidence from this week: current stored values from Day 2, typed and converted values from Day 3, and boundary comparisons from Day 4. Today the team combines separate evidence checks into a simple go/no-go readiness expression.

The problem today is Boolean combination, not branch routing. You will predict whether combined conditions are `True` or `False`, explain which part controls the result, and name whether your support need is state, type, boundary, or Boolean reasoning.

Engineering task	Decide whether a pump record is ready for supervisor review using separate evidence checks that must all be traceable.
Computational move	Combine condition results with `and`, `or`, and `not` while preserving the reason each condition is true or false.
Python focus	Boolean values, comparison results, `and`, `or`, `not`, parentheses, and compound expression explanation.
Evidence to keep	typed value, boundary result, label check, combined truth value, controlling condition, and support category.

Day 5 Boolean rules

1. `A and B` is `True` only when both parts are `True`.
2. `A or B` is `True` when at least one part is `True`.
3. `not A` reverses a Boolean result.
4. Parentheses make the intended grouping visible.
5. A reliable go/no-go note names the separate evidence checks before naming the combined result.

1 Read separate evidence before combining it

recognize predict

Use the week-one pump record below. Each named condition stores one separate evidence result.

```
# Typed values and boundary limits
flow_lpm = 20.1
low_limit_lpm = 19.6
high_limit_lpm = 20.4
approval_label = "reviewed"
operator_initials = "AM"
retry_count = 0
sensor_ready = True

# Separate evidence checks
lower_ok = flow_lpm >= low_limit_lpm
upper_ok = flow_lpm <= high_limit_lpm
flow_ok = lower_ok and upper_ok
approval_ok = approval_label == "reviewed"
setup_ok = sensor_ready and operator_initials != ""
no_retry_needed = retry_count == 0
record_ready = flow_ok and approval_ok and setup_ok and no_retry_needed
```

Pts.	Prompt	My evidence
1	Which two comparisons create the separate lower and upper boundary checks?	
1	Which condition checks the text label rather than numeric flow?	
1	Which condition checks whether a retry is needed?	
1	Which final expression combines all readiness evidence?	

2 Trace an `and` combination from separate evidence

trace explain

Complete the table. Do not skip directly to the final result; show each part.

Pts.	Combined condition	Left part	Right part	Combined result and reason
2	`lower_ok and upper_ok`			
2	`sensor_ready and operator_initials != ""`			
2	`flow_ok and approval_ok`			
2	`setup_ok and no_retry_needed`			

State/type/boundary connection

A Boolean expression is only as trustworthy as the evidence pieces inside it. If a part came from a string label, a converted number, or a boundary comparison, keep that source visible in your reason.

3 Use `or` and `not` for follow-up evidence

predict | explain

A record may need follow-up if at least one risk is present. Complete the table and explain which part controls the result.

```
needs_flow_review = not flow_ok
needs_approval_review = not approval_ok
needs_setup_review = not setup_ok
needs_followup = needs_flow_review or needs_approval_review or needs_setup_review
```

Pts.	Expression	True or False?	Reason
2	`not flow_ok`		
2	`not approval_ok`		
2	`needs_flow_review or needs_approval_review`		
2	`needs_followup`		

Common trap to check

`or` does not mean every risk is present. It means at least one part is `True`. A good note says which part made the expression true.

4 Build a Week 1 condition table

trace transfer

For each case, use Day 3 type/label evidence and Day 4 boundary evidence before deciding the combined result.

Pts.	Flow	Approval label	`flow_ok`	`approval_ok`	`setup_ok`	`record_ready` and reason
2	`20.1`	`"reviewed"`				
2	`20.5`	`"reviewed"`				
2	`20.0`	`"pending"`				
2	`19.6`	`"reviewed"` with blank initials				

Bridge to Week 2

Week 2 will use these Boolean results to choose routes with `if/else`. Day 5 stops one step earlier: decide the truth value and explain the evidence, but do not write a branch yet.

5 Debug a Boolean readiness claim

debug test explain

A teammate writes this note after seeing that `approval_label` stores a non-empty string:

Readiness note to debug

The record is ready because the flow is in range or the approval label has text.

Pts.	Prompt	My response
2	What Day 3 mistake appears in the teammate's note?	
2	What Day 4 boundary evidence is still needed before trusting `flow_ok`?	
2	Why is `or` unsafe if the engineering rule requires both flow evidence and approval evidence?	
2	Rewrite the readiness note so it names the separate evidence checks and uses `and` correctly.	

Week 1 synthesis habit

When a combined expression surprises you, diagnose the support need by source: current state from Day 2, type/cast/truthiness from Day 3, boundary comparison from Day 4, or Boolean connector from Day 5.

6 Mastery check and support route

transfer | explain

Use your own evidence from this workbooklet. Do not write a generic answer.

Pts.	Prompt	My response
2	Give one example where an `and` expression is false because one required evidence check failed.	
2	In one or two sentences, explain how Day 2 state, Day 3 type evidence, and Day 4 boundary evidence work together in Day 5.	

My mastery check

Mark the strongest statement that is true after this workbooklet.

[ ]	Secure: I can combine separate evidence checks with `and`, `or`, and `not`, and explain which part controls the result.
[ ]	Developing: I can evaluate some combined conditions, but I still need support identifying which evidence check failed.
[ ]	Needs support: I need a smaller example before I can combine state, type, boundary, and Boolean evidence.

My next support move

My issue is mostly: setup / Python syntax / tracing / state history / type conversion / boundary cases / Boolean connectors / confidence / time.
The first evidence source where I got uncertain was: ________________________________
My next small action is: ask a partner / ask instructor / rebuild the condition table / test one failed condition at a time / try the recovery version.

Workbooklet target: I can combine typed values, boundary checks, approval evidence, and Boolean connectors into a traceable Week 1 go/no-go explanation.